

Alexander J. Elliott

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Video Editing • Robotics • Guitar • Photography • Physical AI • Climbing • Geography

EXPERIENCE

Magna International • Mechatronics CO-OP

Jan - Aug 2025 • Newmarket, ON

- Developed Python and TypeScript tools to automate pre- and post-processing for FEA, multibody dynamics, and inertial analysis, improving simulation efficiency
- Architected and deployed Inertia+, automating inertia calculations and engineering report generation, reducing engineering effort by up to 26%
- Developed an engineering visualization platform enabling collaborative review of simulation results and faster design iteration

Key Skills: Python • TypeScript • Engineering Automation • Web Development • Data Visualization

AME Group • Mechanical CO-OP & QAQC CO-OP

Sept - Dec 2023, May - Aug 2024 • Vancouver, BC

- Automated repetitive AutoCAD and Revit workflows using Python, PyAutoCAD, C# macros, and Revit plugins
- Developed tools to generate schematics and convert AutoCAD details, reducing manual drafting work
- Led BIM improvement projects that standardized Revit workflows across multiple teams

Key Skills: Python • C# • CAD Automation • Process Improvement • Software Integration

PROJECTS

Litter Collecting Drone • UBC Mech 458 Capstone

Sep 2025 – Apr 2026

- Designed and manufactured a quadcopter to autonomously detect and retrieve litter in difficult terrain
- Integrated YOLOv8 on a Raspberry Pi Zero 2 W for onboard litter detection
- Programmed a Pixhawk flight controller and integrated perception, flight control, and mechanical systems

Key Skills: Robotics • Computer Vision • YOLOv8 • UAV • Embedded AI • Systems Integration

Automated Spice Dispensing Robot • UBC Mech 423 Final Project

Jan – Apr 2026

- Developed a desktop robotic dispensing system integrating embedded controls, custom electronics, sensing, and desktop software
- Created a voice-enabled C# WinForms interface implementing advanced controls, with a focus on user convenience
- Implemented embedded controls on an MSP430 to drive a stepper motor, enable closed-loop DC motor positioning, and manage feedback from an IR sensor and limit switch
- Designed the mechanical dispensing mechanism, inspired by CNC tool changers

Key Skills: Embedded Systems • Robotics • Motor Control • PCB Design • Sensors • C • C#

LapSim Optimizer • UBC Formula Electric

May – Aug 2024

- Developed a MATLAB lap-time simulation tool using vehicle dynamics models to predict competition performance
- Used simulation results to evaluate powertrain and aerodynamic design trade-offs

Key Skills: MATLAB • Simulation • Numerical Modelling • Data Analysis

EDUCATION

University of British Columbia

2021 - 2026 • Vancouver, BC

BASc Mechanical Engineering • Mechatronics Option • w/ Co-Op • w/ Distinction • 5x Dean's List

SKILLS

Programming: Python • C • C++ • C# • Java • JavaScript/TypeScript • MATLAB • SQL

Engineering Software: SolidWorks • CATIA • Fusion 360 • Onshape • Revit • AutoCAD • MATLAB • Docker

Data & AI: NumPy • Pandas • OpenCV • Computer Vision • TensorFlow

Web Development: React • Three.js • HTML/CSS • Web Applications

Systems: Embedded Systems • Raspberry Pi • Arduino • MSP430 • Docker